Ex 1 $L_A : \mathbb{R}^{(n)} \to \mathbb{R}^{(m)}$ $\qquad \mathrm{rk}\, A = \dim \text{image } L_A = \dim \mathbb{R}^{(n)} - \dim \operatorname{Ker} L_A$
$= n - \dim \operatorname{Ker} L_A$

Ex 2. a) $A \cdot A^{-1} = A^{-1} \cdot A = Id \Rightarrow {}^t(A \cdot A^{-1}) = {}^t(A^{-1}A) = Id$

$\Rightarrow {}^t(A^{-1})\,{}^tA = {}^tA\,{}^t(A^{-1}) = Id \Rightarrow {}^t(A^{-1}) = ({}^tA)^{-1}$

b) $f(Ax, {}^t(A^{-1})y) = f(A^{-1}Ax, y) = f(x,y) = f(x, {}^tA({}^tA)^{-1}y) = f(Ax, ({}^tA)^{-1}y)$

~~as~~ set $z = A^{-1}x$, then $f(z, {}^t(A^{-1})y) = f(z, ({}^tA)^{-1}y)$ so ${}^t(A^{-1}) = ({}^tA)^{-1}$.

Ex 3 a) Pick $w \in W$, we must show ${}^tLw = 0 \Rightarrow w = 0$. Pick w' s.t. $\langle w', w\rangle \neq 0$ $v \in V$ s.t. $w' = Lv$. Then $0 = \langle v, {}^tLw\rangle = \langle Lv, w\rangle = \langle w', w\rangle \neq 0$

b) We must show • $\operatorname{Im}{}^tL \cap \operatorname{Ker} L = 0$ & • $\operatorname{Im}{}^tL + \operatorname{Ker} L = V$

Let $w \in \operatorname{Im}{}^tL \cap \operatorname{Ker} L$, then $w = {}^tLv$ and $Lw = 0$ but

$0 = \langle Lw, v\rangle = \langle L\,{}^tLv, v\rangle = \langle {}^tLv, {}^tLv\rangle \Rightarrow {}^tLv = 0 \Rightarrow v = 0$.

Now $\mathrm{rk}\, L = \mathrm{rk}\, {}^tL$ & so $\dim \operatorname{Im}{}^tL + \dim \operatorname{Ker} L = \mathrm{rk}\, L + \dim V - \mathrm{rk}\, L = \dim V$.

Ex 4 a) Think of $L_A : K^{(n)} \to K^{(r)}$. then $\mathrm{rk}\, A < n$ so by ex 1, $\dim \operatorname{Ker} L_A > 0$

But $V = \operatorname{Ker} L_A$ is the set of solutions to the homogeneous system induced by A.

(NB $\dim V = n - \mathrm{rk}\, A$)

b) Let $L'_A : (K')^{(n)} \to (K')^{(r)}$ be the linear map and $V' = \operatorname{Ker} L'_A$. Then $\dim V' = n - \mathrm{rk}\, A = \dim V$ □

Ex 5 Let E_{ij} be the matrix whose only non zero entry is 1 in the i-th row & j-th column. Then $E_{ij}X$ is such that the rows $(E_{ij}X)_k$ are all 0 except for the i-th row $(E_{ij}X)_i = (x_{j1}, x_{j2}, \dots, x_{jn})$ (I used X instead of M!)

So $Tr(E_{ij}X) = x_{ji} = 0 \quad \forall j, i \Rightarrow X = 0$

Ex 8 We claim that $(N^i)_{kl} = 0$ unless $l \geq k+i$.

Since $N = N^1$ is upper triangular, $(N^1)_{kl} = 0$ unless $l \geq k+1$.

By induction we may assume the statement for i.

$N^{i+1} = N^i N$ so $(N^{i+1})_{kl} = \sum (N^i)_{kt} N_{tl}$. If $l < k+i+1$ then either $t < k+i$ or $l < t+1$. (Otherwise $t \geq k+i$ & $l \geq t+1$ so $l \geq k+i+1$!) & if $l < t+1$ $N_{tl} = 0$ □

~~But if $t < k+i$ then $(N^i)_{kt} = 0$~~

$\underline{Ex\ 9}$ We have $0 \neq \operatorname{Ker} T \subset \operatorname{Ker} T^2 \subset \dots \subset \operatorname{Ker} T^m = E$

Let $v_1 \dots v_{i_1}$ be a basis of $\operatorname{Ker} T$, $v_1 \dots v_{i_1}, v_{i_1+1} \dots v_{i_2}$ a basis of $\operatorname{Ker} T^2$ and so on.

We have $T(\operatorname{Ker} T^i) \subset \operatorname{Ker} T^{i-1}$ $\left(\text{as if } T^i w = 0, \text{ then } T^{i-1} T w = 0 \text{ so } Tw \in \operatorname{Ker} T^{i-1} \right)$

In this basis Tv_j is a linear combination of $v_1 \dots v_{j-1}$ & so the corresponding matrix is strictly upper triangular.

$\underline{Ex\ 10}$ $(I+N)^{-1} = I - N + N^2 - N^3 + N^4 \dots \pm N^m$ if $N^{m+1} = 0$.